

Proof. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be convex, i.e.

$$f(tx + (1-t)y) \leq t f(x) + (1-t) f(y) \quad \forall x, y \in \mathbb{R}, \forall t \in [0, 1]. \quad (1)$$

Let $x_1, \dots, x_n \in \mathbb{R}$ and let non-negative weights $\lambda_1, \dots, \lambda_n$ satisfy $\sum_{i=1}^n \lambda_i = 1$. We prove

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i) \quad (2)$$

by induction on n .

1. Degenerate case. If some index j has $\lambda_j = 1$ and $\lambda_i = 0$ for $i \neq j$, then

$$\sum_{i=1}^n \lambda_i x_i = x_j, \quad \sum_{i=1}^n \lambda_i f(x_i) = f(x_j),$$

so (2) reduces to the identity $f(x_j) \leq f(x_j)$. The situation $\lambda_i = 0$ for all i cannot occur because the weights sum to 1. Hence we may assume that at least two weights are positive.

2. Base case $n = 2$. Put $t := \lambda_1$ (so $1 - t = \lambda_2$). Since $t \in [0, 1]$, (1) yields

$$f(\lambda_1 x_1 + \lambda_2 x_2) \leq \lambda_1 f(x_1) + \lambda_2 f(x_2),$$

which is precisely (2) for $n = 2$.

3. Inductive hypothesis. Assume that for some $k \geq 2$ the statement holds for every choice of k points and weights summing to 1; i.e.

$$\forall \mu_1, \dots, \mu_k \geq 0, \sum_{i=1}^k \mu_i = 1, \forall y_1, \dots, y_k \in \mathbb{R} : \quad f\left(\sum_{i=1}^k \mu_i y_i\right) \leq \sum_{i=1}^k \mu_i f(y_i). \quad (\text{IH})$$

4. Inductive step $k \rightarrow k + 1$. Let $\lambda_1, \dots, \lambda_{k+1} \geq 0$ satisfy $\sum_{i=1}^{k+1} \lambda_i = 1$, and let $x_1, \dots, x_{k+1} \in \mathbb{R}$.

Define

$$\alpha := \sum_{i=1}^k \lambda_i, \quad \beta := \lambda_{k+1} = 1 - \alpha.$$

If $\alpha = 0$ then $\beta = 1$ and we are in the degenerate case of step 1, so assume $\alpha > 0$.

Set the intermediate point

$$y := \frac{1}{\alpha} \sum_{i=1}^k \lambda_i x_i.$$

The coefficients λ_i/α are non-negative and sum to 1, hence y is a convex combination of the first k points.

Apply the convexity inequality (1) to the pair (y, x_{k+1}) with weight α :

$$f(\alpha y + \beta x_{k+1}) \leq \alpha f(y) + \beta f(x_{k+1}). \quad (3)$$

A direct computation shows

$$\begin{aligned} \alpha y + \beta x_{k+1} &= \alpha \left(\frac{1}{\alpha} \sum_{i=1}^k \lambda_i x_i \right) + \beta x_{k+1} \\ &= \sum_{i=1}^k \lambda_i x_i + \lambda_{k+1} x_{k+1} = \sum_{i=1}^{k+1} \lambda_i x_i. \end{aligned} \quad (4)$$

Thus the left-hand side of (3) equals the left-hand side of (2) for $k+1$ points.

By the inductive hypothesis applied to the first k points,

$$f(y) = f\left(\frac{1}{\alpha} \sum_{i=1}^k \lambda_i x_i\right) \leq \frac{1}{\alpha} \sum_{i=1}^k \lambda_i f(x_i). \quad (5)$$

Insert (5) into (3):

$$\begin{aligned} f\left(\sum_{i=1}^{k+1} \lambda_i x_i\right) &\leq \alpha \left(\frac{1}{\alpha} \sum_{i=1}^k \lambda_i f(x_i) \right) + \beta f(x_{k+1}) \\ &= \sum_{i=1}^k \lambda_i f(x_i) + \lambda_{k+1} f(x_{k+1}) \\ &= \sum_{i=1}^{k+1} \lambda_i f(x_i), \end{aligned}$$

which is exactly (2) for $k+1$ points. This completes the induction.

5. Equality condition. For two points, equality in (1) occurs iff either $t \in \{0, 1\}$ (so one weight vanishes) or f is affine on the interval $[x, y]$. During the inductive step, equality in (2) can arise only when

- the convexity inequality (3) is an equality, i.e. either $\alpha \in \{0, 1\}$ (some weight is zero) or f is affine on the segment joining y and x_{k+1} ; and
- the inductive hypothesis (5) is an equality, i.e. f is affine on the convex hull of $\{x_1, \dots, x_k\}$.

Combining these observations yields the following description:

1. If some weight $\lambda_i = 0$, the inequality reduces to a lower-dimensional case;
2. Otherwise, equality holds precisely when f is affine on the convex hull of $\{x_1, \dots, x_n\}$.

In particular, for a strictly convex f the second alternative is impossible, so equality can occur only when a single weight equals 1 and all others are zero.

6. Conclusion. The base case ($n = 2$) and the inductive step prove (2) for every integer $n \geq 1$. Together with the degenerate situation of step1, Jensen's inequality holds for all choices of points and weights, and the equality cases have been completely characterised. $\square$